



rather than pressure. Inanimate objects cannot spoof these and are relatively low-cost, but they may fail if hands are wet, much like a damp touchscreen.

Both torque and capacitive sensors function well under normal conditions, but torque sensors remain more common due to their integration with existing hardware. For now, they are likely to remain the mainstream solution, with capacitive sensors providing added support in select models.

What works and what doesn't

Printed sensors offer two main advantages for in-cabin automotive applications. First, they can be integrated directly into surfaces such as the console or seat fabrics, either by printing or weaving, which allows for a more seamless and less intrusive design. Second, when produced in large volumes, printed sensors have the potential to be significantly more cost-effective, making them attractive to automotive manufacturers. However, despite these benefits, they are not yet widely adopted in commercial vehicles.

Eye movement tracking in vehicles is typically performed using near-infrared cameras. These cameras, which are already integrated into many in-cabin systems, can also track eye movements without requiring additional hardware. This means eye tracking can be implemented as a software-based feature using existing equipment.

EEG-based brain function

Detection Category	3D ToF Camera	In-Cabin Radar
Privacy	Poor	Good
Breathing Detection	Good	Good
Motion Detection	Good	Good
Out of Position	Poor (requires recalibration)	Good
Under Blanket Detection	No	Yes
Sensor Resolution	Medium to High	Medium to Low
Processing Cost	Medium to High	Low
Gesture Control	Yes	Yes
Cost	High	High

Comparison of 3D ToF cameras and radars (Source: In-Cabin Sensing 2025-2035: Technologies, Opportunities, and Markets)

monitoring in cars faces both technical and commercial challenges. Technically, EEG signals are highly sensitive to noise, which poses a major issue in a moving vehicle. Vibrations and head movements introduce artefacts that make the signals difficult to process accurately. Extracting reliable data requires significant processing power and time, while low-quality signals can lead to inaccurate readings. Commercially, EEG devices remain expensive and are primarily used in clinical environments, making them unsuitable for widespread automotive use at present.

Cardiovascular monitoring presents similar difficulties. ECG sensors could be installed on the steering wheel to detect heart rate when the driver is holding it, but vehicle vibrations and inconsistent contact may compromise signal quality. A more practical solution is

radar-based monitoring. Millimetre-wave radar, utilising the Doppler effect, can detect small movements such as a heartbeat from a distance, enabling heart rate measurements without direct contact or wearable sensors.

Although no current vehicles incorporate ECG or EEG sensors into their in-cabin systems, radar-based solutions have begun to appear. For example, one of Volvo's recent electric vehicles includes an in-cabin radar system that can detect the heart rate of passengers in the second row. This demonstrates how existing radar technology can be adapted for basic health monitoring, even without the use of traditional biomedical sensors.

How AI helps

Artificial intelligence (AI) plays a central role in modern in-cabin sensing systems, especially for driver monitoring features such as drowsiness and distraction detection. These systems rely on a combination of hardware inputs, such as cameras tracking gaze direction, head movement, and other facial features, and software that processes this data. Computer vision algorithms, a subset of AI, analyse these inputs. The algorithms assess the drivers' behaviour to determine whether they

Cost Integration

- **Driver monitoring systems (DMS):** Use infrared sensors, often mandated by regulations. These sensors are relatively low-cost and widely adopted.
- **Occupancy monitoring systems (OMS):** Use more advanced sensors like radar and time-of-flight, which are significantly more expensive (for example, \$30-\$40 per module).
- **Cost trends:** With greater adoption, sensor prices are expected to drop (targeting ~\$20 per module).
- **Current market:** Advanced in-cabin sensors are mostly found in mid- to high-end vehicles due to cost constraints. Adoption in low-end vehicles remains limited.